MACHINE LEARNING – UNSUPERVISED LEARNING

Clustering Algorithms

K-Means Clustering

Kmeans is an unsupervised Machine Learning algorithm that partitions data into K clusters, each represented by a centroid (mean position of its points); points are assigned to their nearest centroid, which is then recalculated –repeated until convergence. It tries to ensure that data points within the same cluster are as similar as possible and those in different clusters are as different as possible.

How it works

- Decide the number of clusters (K).
- Pick K random centroids.
- Assign every data point to the nearest centroid.
- Calculate new centroids.
- Repeat until the centroids no longer move.

Trick:

A teacher has 100 students but doesn't know their abilities. She decides to separate them into **3 groups** based on marks:

- High performers
- Average performers
- Beginners
- The teacher doesn't already know which student belongs where—the algorithm discovers the groups automatically.

Advantages

- Fast
- Easy to understand
- Works well on large datasets

Limitations

- Must specify K beforehand
- Doesn't work well with oddly shaped clusters
- Sensitive to outliers

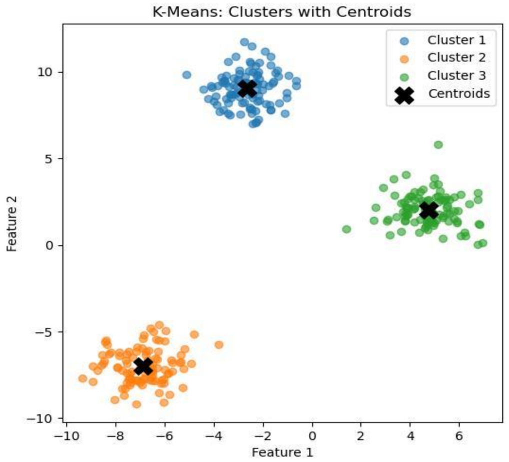
Real-world example

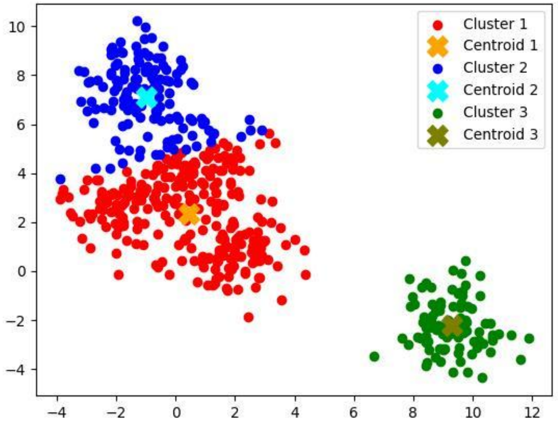
Customer Segmentation: An online retailer groups customers based on Age, Income, Purchase frequency. The business may discover:

- Premium customers
- Regular shoppers
- Occasional buyers

Allows business to run different marketing campaigns for each group.

K Means clusters





Hierarchical Clustering

Hierarchical Clustering creates a tree-like structure (dendrogram) – a tree of nested clusters – by merging (agglomerative) or splitting (divisive) groups based on a linkage criterion, without deciding the number of clusters upfront. Shows how data points are related. Instead of creating all clusters at once, it builds them step by step.

How it works

There are two methods:

Agglomerative (Bottom-Up)

- Every point starts as its own cluster.
- Nearest clusters are merged repeatedly.

Divisive (Top-Down)

- Start with one large cluster.
- Split it into smaller clusters until each point stands alone.

Trick:

Imagine organizing your family tree:

- Individual people
- Families
- Communities
- Cities

Everything is connected in levels

Advantages

- No need to choose K initially
- Easy to visualize relationships
- Good for small datasets

Limitations

- Slow for large datasets
- Difficult to undo wrong merges

Real-world example

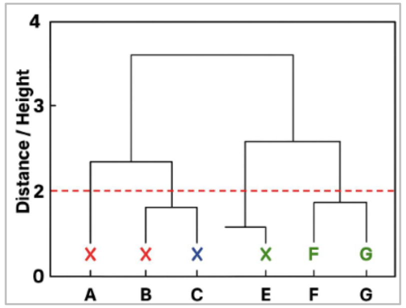
Grouping similar news articles: A news website groups articles into:

- Sports
- Cricket
- IPL
- Team-specific news

Each level becomes more specific

Hierarchical clusters (Dendrograms)

a)



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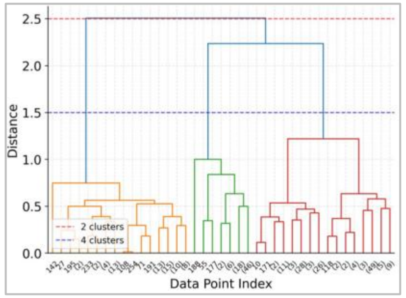

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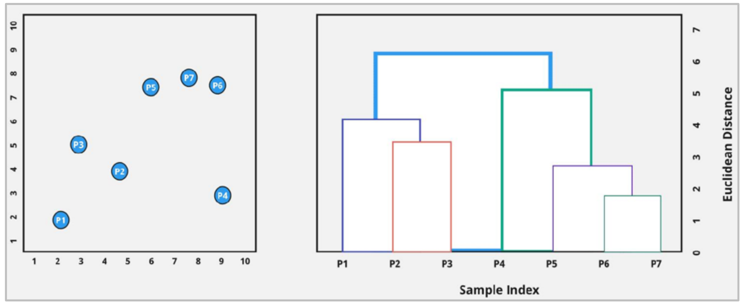


DBSCAN Clustering (Density-Based Spatial Clustering)

DBSCAN is a density based method that groups together points that are closely packed and marks isolated points as noise (outliers). Uses 2 parameters epsilon (neighborhood radius) and minPts (minimum points) – to classify points as core, border or noise

How it works

- For each point, looks at its neighbourhood within a set radius (ε)
- Enough neighbours -> it's a core point, starts/joins a cluster
- Near a core point, but not dense itself -> becomes a border point
- No dense neighbourhood nearby -> labelled as noise

Trick:

Imagine people standing in a park.

- Where many people stand together → one group.
- Someone standing alone → not part of any group.
- DBSCAN naturally ignores isolated people.

Advantages

- Finds clusters of any shape
- Detects outliers
- Doesn't require K

Limitations

- Choosing the right parameters can be difficult
- Less effective when cluster densities vary greatly

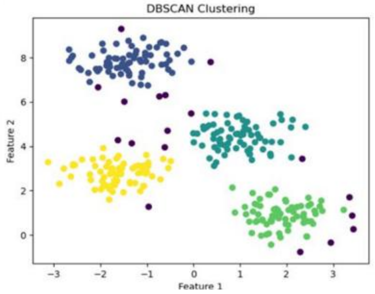
Real-world example

Credit Card Fraud Detection

- Most customer transactions follow normal patterns.
- Rare or unusual transactions appear isolated and can be flagged as potential fraud.

DBSCAN Clusters

a)



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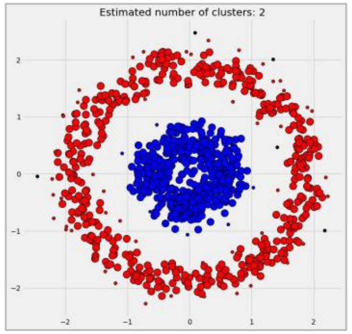

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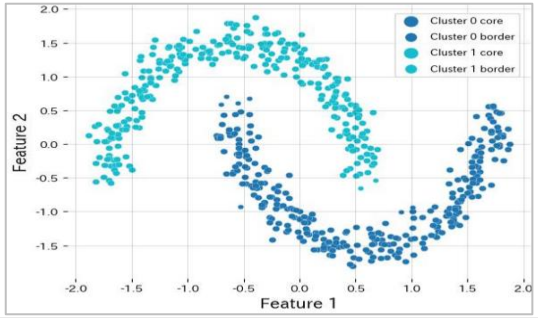


Mean Shift Clustering

Mean Shift identifies clusters by repeatedly moving toward the highest density of nearby data points, eventually settling at density peaks.

How it works

- For each point, looks at nearby points within a window (the bandwidth)
- Shift the point towards the average position of its neighbours
- Repeat until the point stops moving
- Points that converge to the same peak belong to the same cluster

Trick:

Imagine climbing a hill while blindfolded.

- At every step you move in the direction where the ground rises the most.
- Eventually you reach the top of the hill.
- Every hilltop becomes a cluster center.

Advantages

- Automatically finds the number of clusters
- Works well for irregular cluster shapes

Limitations

- Slow on large datasets
- Sensitive to the bandwidth (window size)

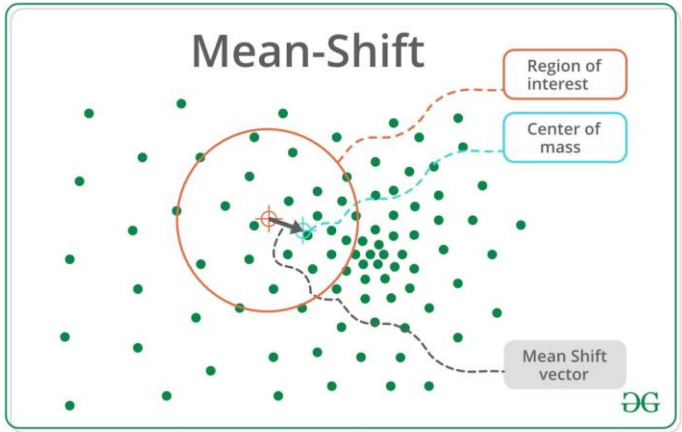
Real-world example

Image Segmentation

- A self-driving car groups similar-colored pixels together to distinguish roads, vehicles, and pedestrians.

Mean Shift Clusters

a)



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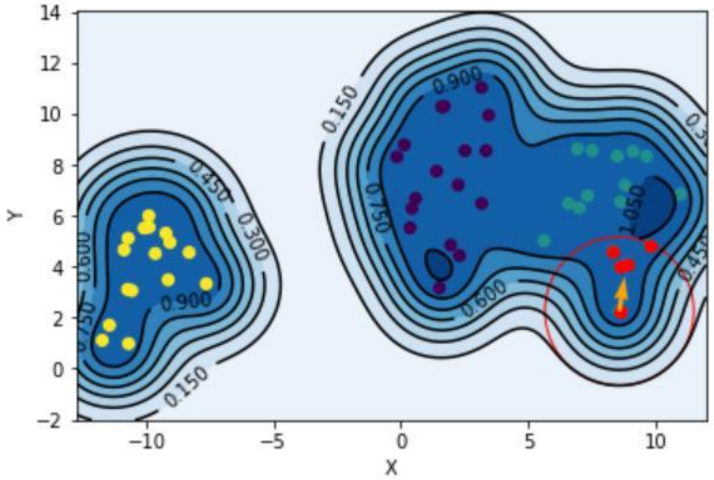


Gaussian Mixture Model (GMM) Clustering

A Gaussian Mixture Model assumes that the data is generated from several Gaussian (bell-shaped) distributions. Each data point has a probability of belonging to every cluster.

How it works

- Start with several Gaussian distributions.
- Estimate the probability that each point belongs to each distribution.
- Update the distributions iteratively using the Expectation-Maximization (EM) algorithm.
- Stop when the model stabilizes.

Trick:

A person may enjoy:

- 70% Action movies
- 30% Comedy movies

Instead of forcing that person into only one category, GMM allows partial membership.

Advantages

- Handles overlapping clusters
- More flexible than K-Means

Limitations

- Computationally slower
- Assumes data follows Gaussian distributions

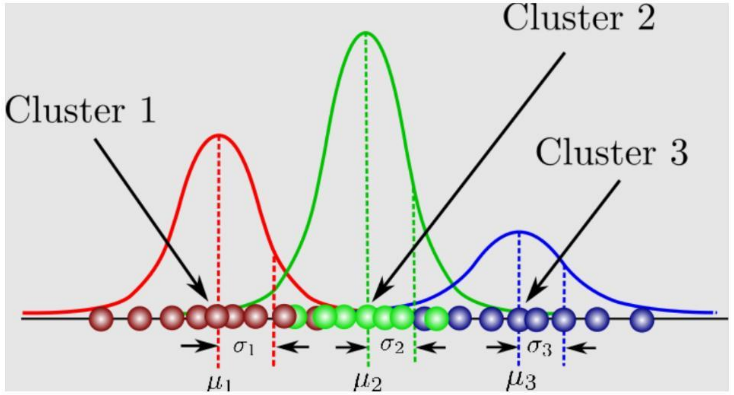
Real-world example

Medical Diagnosis

- A patient's symptoms may partly match multiple diseases. GMM estimates the probability of each diagnosis rather than making a strict assignment

GMM Clusters

a)



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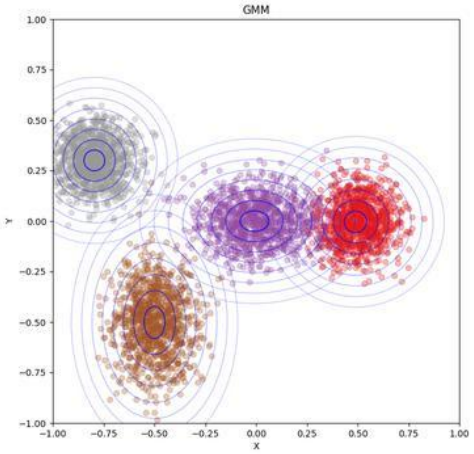


OPTICS Clustering

OPTICS (Ordering Points To Identify the Clustering Structure) is a density-based algorithm that can identify clusters with different densities, addressing one of DBSCAN's main limitations.

How it works

- Orders data points based on density relationships.
- Builds a reachability plot.
- Extracts clusters at multiple density levels.

Trick:

Imagine people gathering at a concert:

- A tightly packed crowd near the stage
- Smaller, more spread-out groups farther away

OPTICS can identify both kinds of groups.

Advantages

- Handles varying densities
- Finds irregularly shaped clusters
- Doesn't require a fixed K

Limitations

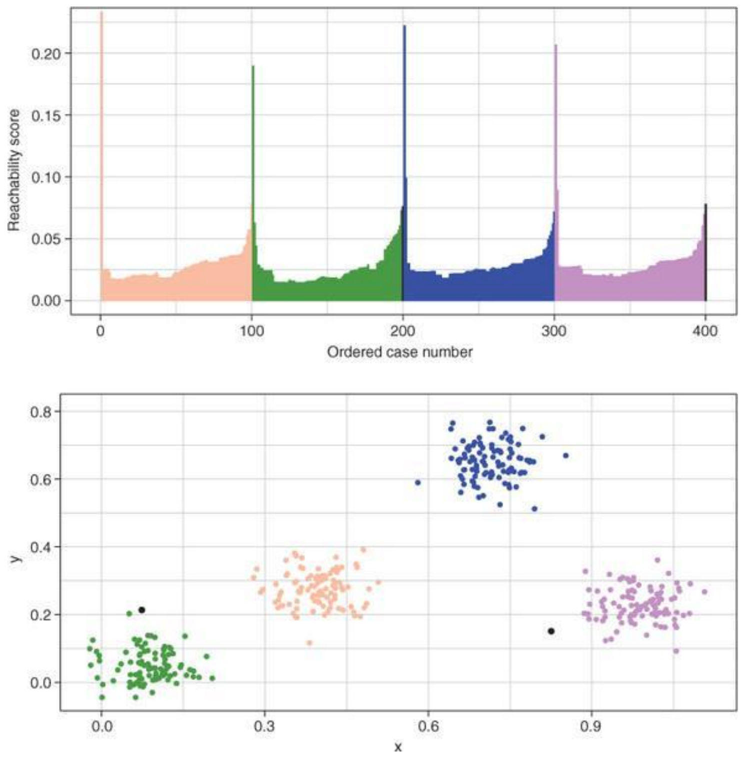
- More complex than DBSCAN
- Harder to interpret

Real-world example

Earthquake Analysis

- Scientists group earthquake epicenters where some regions are very dense and others are more spread out.
- Best for Geospatial analysis, Scientific datasets

OPTICS Clusters



BIRCH Clustering

BIRCH (Balanced Iterative Reducing and Clustering using Hierarchies) is designed for very large datasets. It summarizes the data into a compact tree structure before performing clustering.

How it works

- Build a Clustering Feature (CF) tree.
- Compress similar points.
- Perform clustering on the summarized data

Trick:

Instead of sorting every book individually in a huge library, you first organize books into shelves and sections, then classify them further

Advantages

- Very fast
- Memory efficient
- Scales well to big data

Limitations

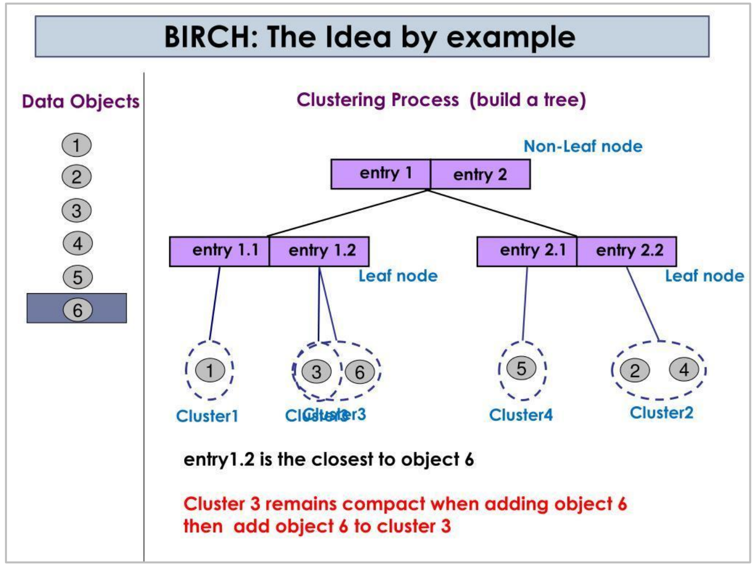
- Best suited to roughly spherical clusters
- Less effective with highly irregular shapes

Real-world example

Telecom Customer Analysis

- A telecom company clusters millions of customer records to identify usage patterns without processing every record repeatedly.

BIRCH Clusters



Spectral Clustering

Spectral Clustering converts the dataset into a graph of connected points and uses linear algebra (eigenvalues and eigenvectors) to partition the graph into clusters.

How it works

- Build a similarity graph.
- Compute the graph's Laplacian matrix.
- Use eigenvectors to embed the data in a lower-dimensional space.
- Cluster the embedded points (often with K-Means).

Trick:

- Imagine a social network. People who interact frequently form communities. Spectral Clustering identifies these communities based on the strength of their connections.

Advantages

- Excellent for complex, non-convex shapes
- Works well when clusters are connected in graph-like structures
-

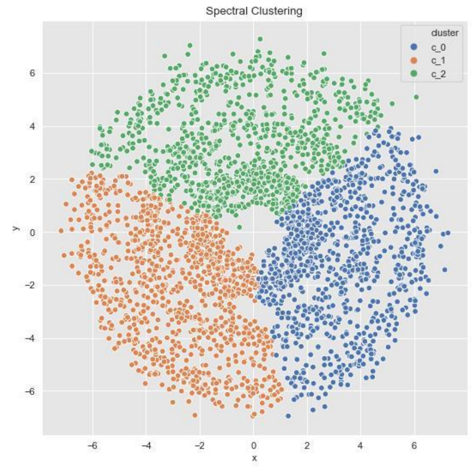
Limitations

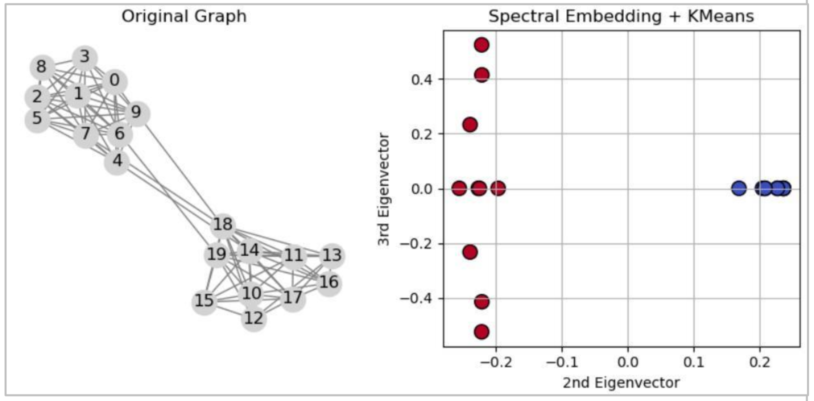
- Computationally intensive
- Not suitable for very large datasets

Real-world example

- **Social Network Community Detection**
- A platform identifies groups of users who interact frequently with each other.

Spectral Clusters





Affinity Propagation Clustering

Affinity Propagation identifies representative data points called exemplars and forms clusters around them, without requiring you to specify the number of clusters.

How it works

- Data points exchange "responsibility" and "availability" messages.
- The most representative points become exemplars.
- Remaining points join the nearest exemplar.

Trick:

- In a group of friends, one person naturally becomes the organizer. Others gather around that person because they are the best representative of the group.

Advantages

- Automatically determines the number of clusters
- Finds representative examples

Limitations

- Computationally expensive
- Not ideal for very large datasets

Real-world example

Product Recommendations

- An e-commerce platform groups similar products and chooses one representative product to display as the main recommendation

Affinity Propagation Clusters